

Movie Recommendation System

Capstone 8 | Machine Learning Models | Summary Report

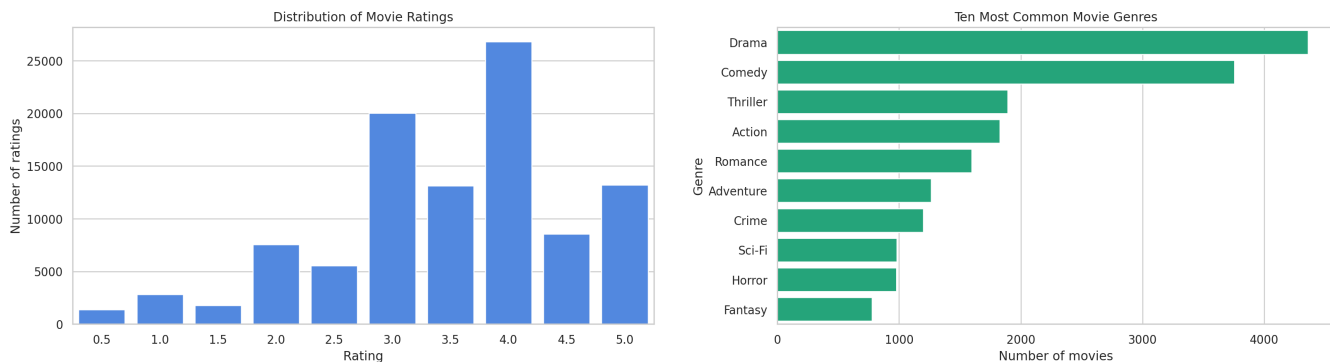
Project objective

This project develops an explainable content-based recommendation system using the MovieLens Latest Small dataset. It recommends movies similar to a selected title, ranks the Top-N results, explains each suggestion through shared genres or tags, and optionally adds a reliable average-rating signal.

Dataset and exploration

Users	Movies	Ratings	Mean rating	Sparsity
610	9,742	100,836	3.502	98.30%

The matrix is highly sparse: most users rated only a small fraction of the catalogue. A content-based model can still recommend an item when genre or tag metadata is available. Drama and Comedy are the most common genres, and ratings are concentrated around 3-4 stars.



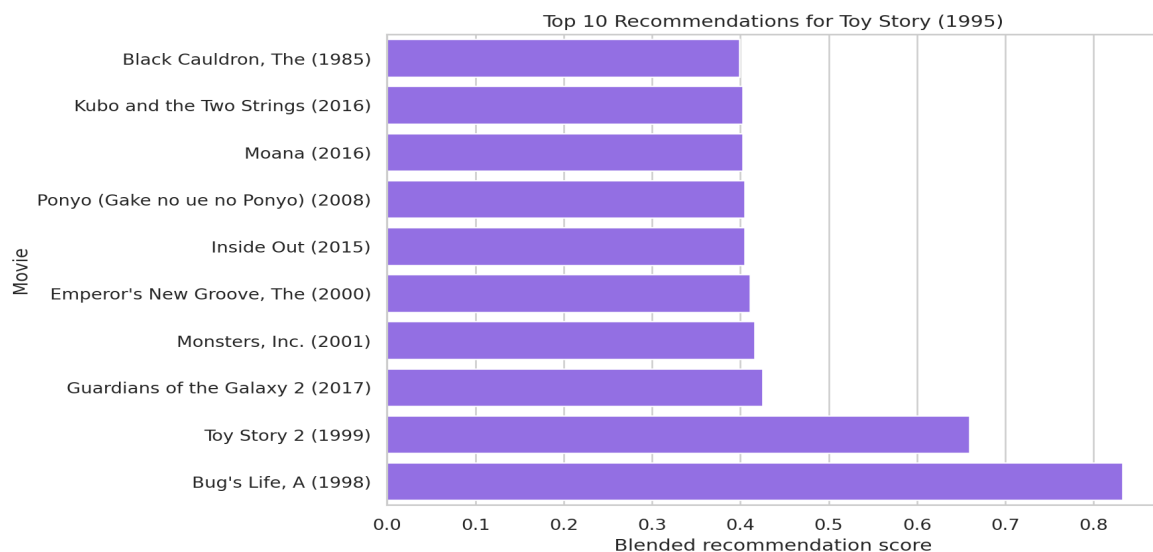
Method

1. Clean movie genres and user tags. 2. Join all text for each movie. 3. Convert text to TF-IDF vectors. 4. calculate cosine similarity between the selected movie and every candidate. 5. blend 85% content similarity with 15% Bayesian rating quality. 6. rank results and generate shared-feature explanations.

Results: movies similar to Toy Story (1995)

The table shows the five highest-ranked items from the reproducible Top-10 output. Content similarity remains the main signal; the rating component helps prefer candidates with stronger and better-supported ratings.

Rank	Movie	Similarity	Avg rating	Why recommended
1	Bug's Life, A (1998)	0.862	3.52	shared genres: Adventure, Animation, Children, Comedy; shared tags: pixar
2	Toy Story 2 (1999)	0.644	3.86	shared genres: Adventure, Animation, Children, Comedy, Fantasy; shared tags: pixar
3	Guardians of the Galaxy 2 (2017)	0.368	3.93	shared genres: Adventure; shared tags: fun
4	Monsters, Inc. (2001)	0.358	3.87	shared genres: Adventure, Animation, Children, Comedy, Fantasy
5	Emperor's New Groove, The (2000)	0.358	3.72	shared genres: Adventure, Animation, Children, Comedy, Fantasy



Findings and limitations

Findings: The strongest recommendations share several family-animation genres and sometimes the Pixar tag. Each result has a concise, human-readable reason. **Limitations:** Metadata can be incomplete; broad genres may create ties; popularity bias can remain; and this item-to-item model does not fully personalize to a user. **Next steps:** build user profiles from multiple liked movies, add matrix factorization, and evaluate with Precision@K or Recall@K.

Source: GroupLens Research, MovieLens Latest Small - <https://grouplens.org/datasets/movielens/latest/>